Milestone - 3 Report

Introduction:
This milestone includes using a RAG model with a pretrained knowledge database for predictions

Performance:

Model	Validation Loss	Prediction performance (MAP@3)
Baseline (sample_submission.csv)	-	0.3525
TF-IDF	0.29617	0.32294
Word2vec	0.34792	0.30423
sBERT	0.42308	0.38653
BART-MNLI	0.56583	0.49985
DaBERTa (3 epochs)	-	0.62177
RoBERTa (3 epochs)	-	0.49958
DaBERTa (8 epochs at 2e-5)	-	0.74064
DaBERTa (16 epochs at 2e-5)	-	0.71945
DaBERTa (8(2e-5) + 8(5e-6))	-	0.73316
DaBERTa (8(2e-5) + 5(5e-6))	-	0.74397
RAG model (MiniLM-v6)	-	0.68536

Conclusion:
The experiments began with a RAG model using a prebuilt vector database for classification, it produced a worse outcome than the deBERTa runs. Not suitable for this project. Cannot be trained further.